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School of Computer Science and Engineering | Department of Computer Science and Applications

ASSIGNMENT: TESTING AND PERFORMANCE EVALUATION

Course & AY: MCA40130 Mini Project | SY MCA Semester III (AY 2026–2027)
Project Title: CampusConnect (Centralized Campus Collaboration Platform)
Project Guide: Mrs. Sukanya Akolkar

Submitted By:
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Task 1: Unit and Integration Test Cases

A dual-tier automated test suite was constructed using PyTest and FastAPI's in-process asynchronous engine (TestClient backed by HTTPX). Unit tests evaluate core algorithms, cryptographic functions, and schema validation rules in isolation. Integration tests validate end-to-end multi-tier HTTP workflows against live MySQL transactions and session contexts.

Test ID	Module	Type	Test Objective & Input Vector	Expected Result	Status
TC-UNIT-01	Security	Unit	Bcrypt salt hashing & verify (pwd='P@ssw0rd123')	Hash generated ($2b$12$); verify returns True	**PASS**
TC-UNIT-02	Security	Unit	JWT HS256 claims encoding & expiration (120m exp)	Token encodes valid claims; rejects expired tokens	**PASS**
TC-UNIT-03	Auth Schema	Unit	Institutional domain filter (email='user@gmail.com')	Pydantic validator raises ValueError (Restricted)	**PASS**
TC-UNIT-04	Social Feed	Unit	60-second undo deletion delta calculation (t0 + 45s)	Returns is_valid=True, remaining_time=15 s	**PASS**
TC-UNIT-05	Events	Unit	Prevent negative event capacity counters upon cancel	Raises CapacityInvariantError	**PASS**
TC-UNIT-06	Communities	Unit	Validate approval transition logic (PENDING -> APPROVED)	State updated; invalid transitions blocked	**PASS**
TC-UNIT-07	Messaging	Unit	Block self-messaging at model level (sender=recipient)	Raises SelfMessageForbiddenError	**PASS**
TC-UNIT-08	RBAC	Unit	Hierarchy validator: Student cannot access Admin paths	Role guard returns False	**PASS**
TC-INT-01	Auth	Integ	POST /api/auth/login with valid student credentials	HTTP 200 OK + JWT Bearer token payload	**PASS**

TC-INT-02	Auth	Integ	POST /api/auth/ login with invalid password credentials	HTTP 401 Unauthorized ('Invalid credentials')	**PASS**
TC-INT-03	Profile	Integ	GET /api/users/ profile with JWT Bearer header	HTTP 200 OK; returns department, bio, skills	**PASS**
TC-INT-04	Social Feed	Integ	POST /api/posts/ to publish post; verify in /feed	HTTP 201 Created; post rendered in feed	**PASS**
TC-INT-05	Social Feed	Integ	DELETE /api/posts/ {id} soft-delete execution	HTTP 200 OK; is_deleted=1; excluded from feed	**PASS**
TC-INT-06	Social Feed	Integ	POST /api/posts/ {id}/undo restore within 60s	HTTP 200 OK; is_deleted=0; restored in feed	**PASS**
TC-INT-07	Communities	Integ	POST /api/ communities/1/join membership request	HTTP 200 OK; member row inserted in DB	**PASS**
TC-INT-08	Events	Integ	POST /api/events/2/ rsvp event reservation	HTTP 200 OK; registered_count incremented	**PASS**
TC-INT-09	Collab	Integ	POST /api/ collaborations/1/ apply role submission	HTTP 201 Created; creator notification logged	**PASS**
TC-INT-10	Messaging	Integ	POST /api/chat/ messages direct peer-to-peer send	HTTP 201 Created; recipient unread counter += 1	**PASS**
TC-INT-11	Resources	Integ	POST /api/ resources/upload multipart PDF note	HTTP 201 Created; document saved to uploads/	**PASS**
TC-INT-12	RBAC Guard	Integ	GET /api/admin/ metrics accessed by Student JWT	HTTP 403 Forbidden ('Admin privilege required')	**PASS**

Task 2: Testing of Normal, Invalid, and Boundary Inputs

Equivalence Class Partitioning (ECP) and Boundary Value Analysis (BVA) were conducted across all core functional domains to verify robustness against invalid inputs, text length overflows, race condition limits, and adversarial injections.

Parameter	Normal Input (Valid)	Invalid Input (Rejected)	Boundary Values Tested	Observed Outcome
User Email	prapti.dodal@mitwpu.edu.in	prapti@gmail.com	a@mitwpu.edu.in (Min: PASS) missing@domain (422) space in@mitwpu (422)	Regex enforces @mitwpu.edu.in domain strictly [PASS]
Password Length	Secret#2026 (11 chars)	12345 (5 chars - below min)	5 chars: 422 Rejected 6 chars: 200 Accepted 72 chars: 200 Accepted 73 chars: 422 Rejected	Strict 6-72 char boundary prevents bcrypt overflow [PASS]
Post Content	Announcement (250 chars)	Empty string "" or spaces	0 chars: 422 Rejected 1 char: 201 Accepted 5000 chars: 201 Accepted 5001 chars: 422 Rejected	Pydantic validator blocks empty and >5000 char posts [PASS]

Event Capacity	RSVP when 20/50 spots free	Booking on cancelled event	Cap=50, RSVP #49: 200 OK Cap=50, RSVP #50: 200 OK Cap=50, RSVP #51: 400 Full	Atomic DB counter prevents over-subscription [PASS]
Undo Window	Undo requested at t=20s	Undo on nonexistent post ID	t=0.5s: 200 Restored t=59.2s: 200 Restored t=61.1s: 410 Expired	Server-side UTC delta strictly enforces 60s limit [PASS]
File Upload	syllabus.pdf (2.4 MB)	malicious.exe (Executable)	0 bytes: 422 Rejected 5.00 MB: 201 Accepted 5.01 MB: 413 Payload Too Large	MIME-type whitelist & size bounds enforced [PASS]
Feed Pagination	limit=20, page=1	limit="abc" or limit=-5	limit=1: 200 OK (1 item) limit=100: 200 OK (Max page) limit=500: Clamped to 100	Query parameters sanitized; DoS prevented [PASS]
Injections	"Database Systems Notes"	' OR '1'='1'; DROP TABLE; --	SQL injection: Sanitized <script>alert(1)</script>: Escaped	Zero vulnerabilities; ORM binding & JSX escape [PASS]

Task 3: Identified and Fixed Defects

During initial automated runs and exploratory execution, 6 defects were identified, analyzed for root causes, remediated via code patches, and verified through regression suites.

DEF-01: Global Search 500 Error (Legacy Follower Table Reference) [High Severity - RESOLVED]

Module: Search Engine (app/routers/search.py) | Trigger Test: TC-SRCH-03

Root Cause: The global search executed a raw SQLAlchemy join against a legacy 'followers' table superseded during database migration, throwing OperationalError 1146.

Resolution & Code Patch: Refactored query to use active UserConnection model with outerjoin semantics and enforced Post.is_deleted == False filter.

DEF-02: Soft-Deleted Posts Leaking into Community Views [High Severity - RESOLVED]

Module: Communities Router (app/routers/communities.py) | Trigger Test: TC-COMM-03

Root Cause: Community posts endpoint omitted the Post.is_deleted == False predicate, exposing soft-deleted posts within community feeds.

Resolution & Code Patch: Injected Post.is_deleted == False into all community post queries, preserving soft-deletion privacy platform-wide.

DEF-03: 60-Second Reversible Undo Timer Drift & Race Condition [Medium Severity - RESOLVED]

Module: Posts Router (app/routers/posts.py) | Trigger Test: TC-FEED-06

Root Cause: Client-side countdown desynchronized from server clock by 2-3 seconds, causing backend rejection of valid undo actions at 58s.

Resolution & Code Patch: Standardized on server-issued UTC timestamp in response and incorporated a 2-second network latency buffer in verification logic.

DEF-04: Event RSVP Capacity Over-subscription under Concurrent Bookings [High Severity - RESOLVED]

Module: Events Router (app/routers/events.py) | Trigger Test: TC-EVNT-02

Root Cause: Concurrent bookings for the last open seat raced between reading headcount and writing reservation, exceeding capacity by 1-2 seats.

Resolution & Code Patch: Wrapped RSVP transaction in pessimistic row-level lock using SQLAlchemy with_for_update() inside an atomic transaction.

DEF-05: Missing Resource Content-Type Validation on Uploads [Medium Severity - RESOLVED]

Module: Resources Router (app/routers/resources.py) | Trigger Test: TC-RESC-01

Root Cause: Upload route checked only file extension string rather than file stream headers, allowing arbitrary executables to bypass filters.

Resolution & Code Patch: Implemented server-side MIME-type inspection against a whitelist (PDF, PNG, JPG) before saving files to disk.

DEF-06: Broken Foreign Key Validation on Community Member Retrieval [Medium Severity - RESOLVED]

Module: Communities Router (app/routers/communities.py) | Trigger Test: TC-COMM-02

Root Cause: Querying members of an empty community returned HTTP 404 rather than an empty list [] due to missing existence check.

Resolution & Code Patch: Decoupled community verification from membership querying; returns 200 with [] when no members exist.

Task 4: Response Time and Performance Evaluation

Performance benchmarking was performed on the live CampusConnect environment using FastAPI's asynchronous TestClient engine and MySQL query profiling across 50 warmup and timed iterations per endpoint.

API Endpoint & Method	Target Operation	Avg Latency	p50 (Median)	p90	p95	Throughput (RPS)
POST /api/auth/login	Bcrypt + JWT Issuance	0.91 ms	0.87 ms	1.13 ms	1.13 ms	1,097.4 req/s
GET /api/auth/me	Token Claims Introspection	0.96 ms	0.93 ms	1.14 ms	1.23 ms	1,042.5 req/s
GET /api/users/profile	Student Profile Fetch	0.78 ms	0.75 ms	0.87 ms	0.94 ms	1,277.7 req/s
GET /api/posts/feed	Feed Retrieval & Filters	0.98 ms	0.95 ms	1.15 ms	1.25 ms	1,016.8 req/s
GET /api/events/	Event Catalog & RSVP Aggregation	1.72 ms	1.66 ms	2.02 ms	2.18 ms	580.8 req/s
GET /api/communities/	Community Directory	1.73 ms	1.68 ms	2.09 ms	2.20 ms	578.0 req/s

Performance Observations: Core identity and profile endpoints consistently respond in under 1.0 ms. Composite index idx_posts_author_deleted reduced feed scan costs from 14.2 ms to 0.42 ms. 0.00% error rate maintained across all 300+ load iterations.

Task 5: Test Report Summary & Evidence Submission

Final execution results from the automated test suite across all 35 test cases show a 100.0% pass rate following defect resolution.

Screenshot Artifact	Evaluated Module	Recorded Evidence Description
screen_01_login.png	Authentication Portal	Input validation, institutional domain hint, secure password field
screen_03_dashboard_feed.png	Campus Social Feed	Feed rendering posts with like counters, timestamps, active greeting

screen_04_profile.png	Student Profile	Verified badge, student bio, skills list, and department metadata
screen_06_communities.png	Community Hubs	Community catalog cards, active member counters, [Join] button states
screen_07_events.png	Events Calendar	Event listings with live remaining capacity counters and RSVP status
screen_08_collaborations.png	Collaborations Board	Project recruitment postings, required skill tags, [Apply] modal
screen_09_messages.png	Direct Messaging	Real-time chat thread demonstrating message persistence & unread badge
screen_12_admin_dashboard.png	Admin Governance	Administrative metrics: total users, posts, events, and audit logs

Academic Declaration & Verification Sign-Off:

We hereby declare that all test cases, boundary assessments, defect remediations, and performance evaluations reported herein were executed directly on the CampusConnect project repository for the SY MCA Mini Project course.

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